

WASP-43b

R-2588

Benchmark run · workflow W8-benchmark-deep · record deep-bench_WASP-43_140222 · created 2026-08-27T22:58:16+00:00

BENCHMARK: ACCURATE — fit consistent with published values

Results

Mid-Transit Time (Tmid)	2456710.84523 +/- 0.0007 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.194 +/- 0.043
Transit depth (Rp/Rs)^2	3.77 +/- 1.67 [%]
Orbital Inclination (inc)	78.62 +/- 1.92
Airmass coefficient 1 (a1)	0.951 +/- 0.014
Airmass coefficient 2 (a2)	0.034 +/- 0.022
Scatter in the residuals of the lightcurve fit is	1.45 %
Best Comparison Star	#3 - [579, 160]
Optimal Aperture	3.66
Optimal Annulus	9.2
Transit Duration (day)	0.035 +/- 0.015

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R $\blacksquare$ fit vs published	0.194 $\pm$ 0.043 vs 0.1594 (deviation 0.8 σ)
Duration fit vs published	0.035 d vs 0.0512 d (deviation 1.08 σ)
Mid-time O-C	-1.9 min
Depth SNR	4.5
Residual scatter	1.45 %

Light curve

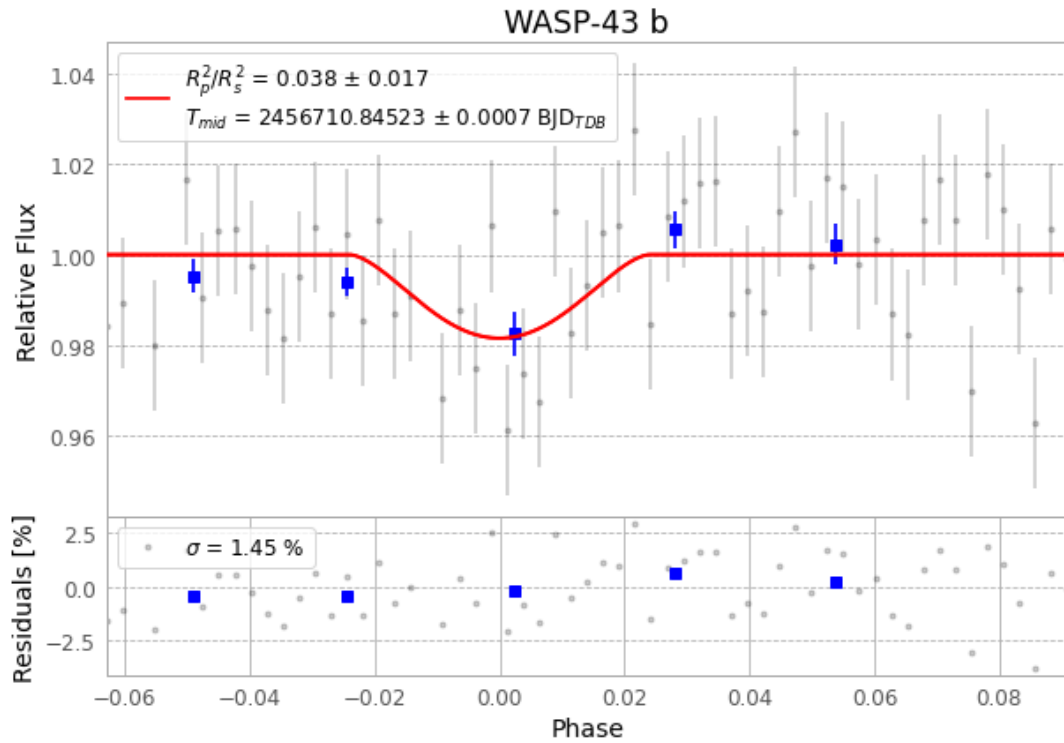


Figure 1 — FinalLightCurve_WASP-43 b_2014-02-21.png (SHA-256 b8fbe2a4bb52cbfa...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [302, 248], 3 comparison star(s). Exact configuration: parameters.inits_json in record.json (SHA-256 8dd84f299985368a...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit a4253ce

Data provenance

61 FITS frames (40 MB), WASP-43140222065415.FITS → WASP-43140222095412.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-WASP-43-140222.log (43c3c5b7e018...), FinalLightCurve_WASP-43 b_2014-02-21.png (b8fbe2a4bb52...), FinalLightCurve_WASP-43 b_2014-02-21.csv (8e076dcc08fe...)

Compute resources

Wall time 97.8 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-08-27 22:58Z.